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الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM105
Introduction to Computer Programming

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 21/April/2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 1.5 hours

Instructions:

- Total Marks: **60**
- This exam consists of 5 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1 [6 Marks]

For each question, choose the most appropriate answer. Copy the answer to the External Answer booklet. Each question carries 2 marks.

1. Which statement correctly declares and initializes a character variable named ch?
   a. `char ch = "x";`
   b. `char ch = 'x';`
   c. `ch char = 'x';`
   d. `character ch = 'x';`
2. What is printed by the statement: `System.out.println(10 % 4);`
   a. 2.5
   b. 2.0
   c. 2
   d. 4
3. Which of the following keywords is used to make a decision in Java?
   a. `for`
   b. `if`
   c. `print`
   d. `printf`

Question 2 [12 marks]

The following code includes some errors. Find the errors and rewrite the code after correction.

```
public class 2SimpleCode {
    public static main(String[] args) {
        int n = "5";
        int sum = 0;
        for (i = 1, i <= n, i++)
            sum = sum + i;
        System.out.println("Sum is: " , sum);
    }
}
```

Question 3 [15 Marks]

Trace the following pieces of code and write their exact output:

a.

```
int x = 5;
int y = 4;
System.out.println(y / 6);
System.out.println(x % 6);
System.out.println(x * y + y * 2 );
System.out.println(x * (y + y) * 2 );
```

b.

```
int i = 3;
if (i % 2 != 0 || i>= 10) {
   System.out.println(i);
   i += 3;
   System.out.println(i); }
else {
   System.out.println(i);
   i += 2;
   System.out.println(i); }
```


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c.

```
String s = "Negative";
System.out.println(s.charAt(0));
if (s.charAt(0)=='n')
   System.out.println("N");
else
   System.out.println("P");
if (s.equals("negative"))
   System.out.println("OK");
```

d.

```
int i;
for (i = 1; i <= 7; i++) {
   if (i > 4)
       System.out.println(i);
}
```

For questions 4 and 5, copy the below code into your answer sheet and continue the rest of the code according to the given specifications of each question.

```
import java.util.Scanner;
public class MTA {
      public static void main(String[] args) {

   }
}
```

Question 4 [20 Marks]:

The monthly household electricity consumption must not exceed 600 kWh. The consumption level is classified according to the following table:

Consumption Level	Electricity Consumption (kWh)
Low	usage ≤ 120
Medium	120 < usage ≤ 300
High	300 < usage ≤ 480
Very High	480 < usage ≤ 600

You need to write a Java program that works as follows:

- Read an integer value from the user representing the number of kilowatt-hours (kWh) consumed in a month.
- Determine and print the corresponding electricity consumption level based on the table above.
- If the user enters a value that is negative or greater than 600, the program should print an appropriate error message.

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Here are different samples of program run to help you understand how the program works:

1	Enter the number of kilowatt-hours consumed: 80 Low
2	Enter the number of kilowatt-hours consumed: 220 Medium
3	Enter the number of kilowatt-hours consumed: 480 High
4	Enter the number of kilowatt-hours consumed: 580 Very High
5	Enter the number of kilowatt-hours consumed: 700 Invalid!

Question 5 [7 Marks]:

Write a for loop to print the multiples of 6 from 6 to 60 in one line.

The output should be exactly as follows:

6 12 18 24 30 36 42 48 54 60

End of questions

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